1 Linear Actuator Options

This document covers the types of linear actuators considered for the front oscillation and stabilization of the UAV launch system. Multiple types were considered, ranging from prebuilt purchased models to possible self configurable options. The main factors considered are cost, linear force, linear speed, controllability and accuracy. A rotational force at the base of the launcher was also considered.

1.1 Motor With Lead Screw or Ball Screw

A motor can be connect to a lead screw to transfer rotation motion and force into linear. This can be done using differing motors that are explored in detail in section 2. The main two mechanisms to cause this conversion is a lead screw or a balls screw. They both work on similar principals using a motor and gear box to rotate a screw forward and backwards.

The lead screw uses a threaded shaft and nut mechanism to convert the motion, a ball screw instead use a threaded shaft and ball nut to convert the motion. Ball screws in turn have significantly lower friction loss. The advantages and disadvantages of both are shown below.

1.1.1 Lead Screw Advantages

- Low cost.
- High mechanical advantage.
- High linear force can be achieved.
- Simple mechanical construction.
- Relatively simple control system.
- A lead screw can be self-locking depending on its lead angle.
- A self-locking screw can help hold a vertical load without continuously powering the motor.
- Good for vertical loads to remain stationary.
- Less sensitive to contamination.

1.1.2 Lead Screw Disadvantages

- Lower efficiency than a ball screw.
- Higher friction and heat generation.
- Lower maximum speed than many ball-screw systems.
- Backlash can reduce positioning accuracy.

- Position feedback is required for accurate positioning.
- High-speed operation can reduce service life.
- Requires more motor torque for similar linear force

1.1.3 Ball Screw Advantages

- Higher efficiency than a lead screw.
- Higher achievable linear speed.
- Lower friction.
- Lower heat generation.
- High force capability.
- Good positioning accuracy when combined with position feedback.
- Suitable for rapid and repeated motion.

1.1.4 Ball Screw Disadvantages

- Higher cost than a lead screw.
- Ball screws can be back-driven by a vertical load.
- A brake or continuous motor torque may be required to hold a vertical load.
- Requires lubrication and maintenance.
- More mechanically complex than a lead screw.
- Position feedback is required for accurate positioning.
- Requires more careful alignment

2 Motors Paired with Lead/Ball Screw

While using a ball or lead screw a DC, Stepper or Servo motor can be used. This document will not be going into detail of the functionality or mechanics or each. Below is a list of their advantage or disadvantages, when connected to either lead or ball screws.

2.1 Dc Motor

2.1.1 DC Advantages

- Relatively low cost.
- Simple electrical control.
- High rotation low speed.
- Direction can be revered

2.1.2 DC Disadvantages

- Does not inherently know position, requires an encoder
- Can over shoot position easily.
- Speed and torque affected by load
- Lead-screw friction reduces efficiency.
- Maximum linear speed is limited.
- Holding position may require continuous motor torque or break.
- may require gear box to achieve required torque.

2.2 Stepper Motor

2.2.1 Stepper Advantages

- Relatively inexpensive
- Simple position commands
- Does not require encoder
- High holding torque
- Can be very smooth and command resolution

2.2.2 Stepper Disadvantages

- Problematic if load excess available torque
- Open-loop can not confirm position
- Torque diminishes as rotation speed increases
- Can generate significant heat while stationary
- May require larger motor to achieve high-speed
- Break or holding mechanism may be necessary for vertical load.

2.3 Servo Motor

2.3.1 Servo Advantages

- Closed-loop position control.
- High positioning accuracy and repeatability.
- The controller can compensate for changes in load.
- Good dynamic performance, rapid acceleration and deceleration.
- Automatically corrects position errors.
- Good torque capability over broad operation range.

2.3.2 Servo Disadvantages

- Higher cost than a basic DC motor system.
- Requires a compatible servo drive/controller
- More complex setup.
- Can overshoot or oscillate without proper tuning.
- Holding torque can generate high heat.
- A mechanical break may still be required to hold load without power.

3 Linear Motor

A linear motor produces linear motion directly through electromagnetic forces rather than converting rotary motion into linear motion through a screw or other mechanical transmission.

3.0.1 Advantages

- Direct generation of linear motion.
- Very high achievable linear speed.
- High acceleration and deceleration.
- High positioning accuracy.
- High repeatability.
- No screw or gearbox backlash.
- Low mechanical wear.
- Excellent dynamic performance.

3.1

subsectionDisadvantages

- Significantly higher cost than conventional actuators.
- Requires sophisticated control electronics.
- Position feedback is typically required.
- Does not inherently provide a self-locking mechanism.
- A brake or continuous control force may be required for vertical loads.
- More difficult to integrate mechanically.
- Generally unnecessary when extremely high speed is not required.

4 Basic Prebuilt Linear Actuator

Basic prebuilt electric linear actuator were explored as a possible option for the front rail actuation. However this was quickly shown to be not feasible due to the slow linear speed of most models making it nearly unusable for the required launch time frame.

5 Rotation Torque Instead of Linear Force

In the application of the launch system a motor mounted at the point axis of rotation is also a feasible solution. This inherently changes some of the required specification need from the motors. Instead of linear speed it would be angular speed as well as applied torque. The general torque would need to be significantly greater due to the force applied at the base.

This option is still in considerations as it the motor may be larger and more expensive but could be compensated with build gear box and no longer the need for a lead or ball screw mechanism

6 Summary

In summary, there are multiply options for actuation of the front end of the UAV launch system. The main leading option being a ball screw with a servo motor.

The lead screw is likely to cause complication with friction at high speeds. The Dc and Stepper motor although cheaper will likely lead to lower position and control accuracy then needed for this application. The linear motor although the best for force and controllability is largely too expensive of an option for the budget of this project. The basic linear actuator is incapable of getting

to required speed to control the actuation properly. The base rotation with a motor and gearbox is a viable solution.

Final part selection Will be completed for both the full size UAV launcher and scaled down version in later documentation, using this documents information along with calculation taken from "UAV Launch Actuator Values".

References

[1] isel USA, "The Differences Between Lead and Ball Screw Actuators." [Online]. Available: https://www.isel-us.com/blog/the-differences-between-lead-and-ball-screw-actuators.

[2] Advanced Motion Controls, "DC Servo Motor." [Online]. Available: https://www.a-m-c.com/dc-servo-motor/.

[3] The Pi Hut, "What's the Difference Between DC, Servo and Stepper Motors." [Online]. Available: https://thepihut.com/blogs/raspberry-pi-tutorials/whats-the-difference-between-dc-servo-amp-stepper-motors.

[4] Tecnotion, "What Is a Linear Motor?" [Online]. Available: https://www.tecnotion.com/faq/what-is-a-linear-motor/.

[5] ACV Group, "Understanding How Electric Actuators Work." [Online]. Available: https://acv-group.com/understanding-how-electric-actuators-work/.